① **Matrix** :- It is a 2-dimensional array.

	col 0	col 1				col 5
R0	(0, 0)	(0, 1)	(0, 2)	(0, 3)	(0, 4)	(0, 5)
R1	(1, 0)	(1, 1)				(1, 5)
R2	(2, 0)	(2, 1)				(2, 5)
R3	(3, 0)	(3, 1)				(3, 5)

- Any problem related to matrix can be solved via following ways:-

1. BFS
2. DFS

3. Using Auxiliary Array, index manipulation, Pointer approach, etc.

- It is important to keep in mind that if using extra S.C, we can reduce T.C then it is a better approach as having less T.C using extra S.C is a better approach than having higher T.C and using constant <u>S.C</u>.

① Whenever you are stuck during interview to reduce T.C, always give a second thought on that "Can we reduce T.C using extra space"? This is one of the trick to reduce T.C

```
① for (int i = 0; i < r; i++)
  {
    for (int j = 0; j < c; j++)
    {
      //do some operⁿ
    } // for j loop
  } // for i loop.
```

$m \times n$ $\quad (m+1)\ (n+1)$

$(i, j) \rightarrow (i-1, j-1)$

Input Array

0	0	0	0	0	0
0	1	0	1	0	0
0	1	0	1	1	1
0	1	1	1	2	2
0	1	0	0	0	0

side = 2

Area = side × side = 2 × 2 = 4